

Roll No.

TME-704

B. TECH. (ME) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

OPERATIONS RESEARCH AND OPTIMIZATION TECHNIQUES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain, how and why operations research methods have been valuable in aiding executive decisions. 10 Marks (CO1)

OR

- (b) Explain the scope and methodology of OR, the main phases of OR and techniques used in solving OR. 10 Marks (CO1)

2. (a) A furniture manufacturer makes two products : chairs and tables. These products are processed using two machines A and B. A chair requires 2 hours on machine A and 6 hours on machine B. A table requires 5 hours on machine A and no time on machine B. There are 16 hours per day available on machine A and 30 hours on machine B. The profit gained by the manufacturer from a chair is ₹ 2 and from a table is ₹ 10. Solve this problem to find the daily production of each of the two products. 10 Marks (CO1)

P. T. O.

OR

- (b) A firm makes two products X and Y and has a total production capacity of 9 tonnes per day. Both X and Y require the same production capacity. The firm has a permanent contract to supply at least 2 tonnes of X and at least 3 tonnes of Y per day to another company. Each tonne of X requires 20 machine hours of production time and each tonne of Y requires 50 machine hours of production time. The daily maximum possible number of machine hours is 360. All of the firm's output can be sold. The profit made is ₹ 80 per tonne of X and ₹ 120 per tonne of Y. Formulate this problem as an LP model and solve it by using graphical method to determine the production schedule that yields the maximum profit.

10 Marks (CO1)

3. (a) Solve the following LP problem using the Simplex method :

Maximize : $Z = x_1 - 3x_2 + 2x_3$

Subject to the constraints :

$$3x_1 - x_2 + 3x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0.$$

10 Marks (CO1)

OR

(b) Use penalty (Big-M) method to solve the following LP problem :

Minimize : $Z = 3x_1 - x_2$

Subject to the constraints :

$$2x_1 + x_2 \geq 2$$

$$x_1 + 3x_2 \leq 3$$

$$x_2 \geq 4$$

$$x_1, x_2 \geq 0.$$

10 Marks (CO1)

4. (a) A steel company has three open hearth furnaces and five rolling mills. The transportation costs (rupees per quintal) for shipping steel from furnaces to rolling mills are given in the following table :

	Rolling Mills						
		M1	M2	M3	M4	M5	Supply
Furnace	F1	4	2	3	2	6	8
	F2	5	4	5	2	1	12
	F3	6	5	4	7	7	14
	Demand	4	4	6	8	8	

What is the Optimal Transportation Cost ?

10 Marks (CO2)

P. T. O.

OR

- (b) The DREAM-DRINK Company has to workout a minimum cost transportation schedule to distribute crates of drinks from three of its factories X, Y and Z to its three warehouses A, B and C. The required particulars are given below. Find the least cost transportation schedule :

10 Marks (CO2)

From/To	A	B	C	Crates Available
X	75	50	50	1040
Y	50	25	75	975
Z	25	125	25	715
Crates Required	1300	910	520	

5. (a) A Computer center has got three expert programmers. The center needs three application programmes to be developed. The head of the computer center, after studying carefully the programmes to be developed estimate the computer time in minutes required by the experts to the application programmes as given in the matrix below. Assign the programmers to the programmes in such a way that the total computer time is least :

10 Marks (CO2)

Programmers	Application Programmes		
	A	B	C
1	120	100	80
2	70	90	110
3	110	140	120

OR

- (b) A company has four machines that are to be used for three jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given in the following table : 10 Marks (CO2)

		Machines			
		W	X	Y	Z
Jobs	A	18	24	28	32
	B	8	13	17	18
	C	10	15	19	22

What are the job assignment pairs that shall minimize the cost ?